

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability theory

Tutorial 1
2026/09/08

1. Show that $E \cap (F \cup G) = (E \cap F) \cup (E \cap G)$.
2. Let E and F be mutually exclusive events in the sample space of an experiment with at least one of $P(E)$ and $P(F)$ being nonzero.
 - (a) Suppose that the experiment is repeated until either event E or event F occurs. What does the sample space of this new super experiment look like?
 - (b) Now we do not stop the experiment even if E or F occurs, and repeat it for infinitely many times. Show that the probability that event E occurs before event F is $\frac{P(E)}{P(E) + P(F)}$.

Hint: Argue that the probability that the original experiment is performed n times and E appears on the n -th time is $P(E) \times (1 - P(E) - P(F))^{n-1}$.

3. Let $\{E_i\}_{i=1}^{\infty}$ be a countable collection of events. Show that

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} P(E_i).$$

This is known as Boole's inequality.

Hint: Define a collection of mutually disjoint events in a suitable way, then apply the countable additivity property of probability.

4. There are n people in a room and each of them owns a hat. Now each person throws his hat into the center of the room, and the hats are then randomly returned to the people so that each person receives one hat. Assume that all possible assignments of hats to people are equally likely.
 - (a) If $n = 2$, what is the probability that no person gets his own hat back?
 - (b) For general n , what is the probability that no person gets his own hat back?